

- (b) A current of 4.0 A flows through a wire of length 1 m and cross-sectional area 1.0 mm^2 , when potential difference of 2 V is applied across its ends.

Calculate the resistivity of the material of the wire.

- (b) A current of 4.0 A flows through a wire of length 1 m and cross-sectional area 1.0 mm^2 , when potential difference of 2 V is applied across its ends.

Calculate the resistivity of the material of the wire.

CBSE 26

2

$$\rho = \frac{RA}{l}$$

$$R = \frac{V}{I} = \frac{2}{4} = \frac{1}{2} \Omega$$

$$\rho = \frac{1 \times 10^{-6}}{2 \times 1}$$

$$= 5 \times 10^{-7} \Omega \text{ m}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1. A wire is gradually stretched to make it 100% longer. Its resistance will increase by :

(A) 100%

(B) 200%

(C) 300%

(D) 400%

1. A wire is gradually stretched to make it 100% longer. Its resistance will increase by :

(A) 100%

(B) 200%

(C) 300%

(D) 400%

CBSE 26

(C) 300 %

1. Two wires of same length and same area of cross-section but made of different material of resistivities ρ_1 and ρ_2 are connected in series. The equivalent resistivity of the combination is :

(A) $\frac{1}{2}(\rho_1 + \rho_2)$

(B) $\rho_1 + \rho_2$

(C) $\sqrt{\rho_1 \rho_2}$

(D) $2(\rho_1 + \rho_2)$

1. Two wires of same length and same area of cross-section but made of different material of resistivities ρ_1 and ρ_2 are connected in series. The equivalent resistivity of the combination is :

(A) $\frac{1}{2}(\rho_1 + \rho_2)$

(B) $\rho_1 + \rho_2$

(C) $\sqrt{\rho_1 \rho_2}$

(D) $2(\rho_1 + \rho_2)$

CBSE 26

(A) $(\rho_1 + \rho_2) / 2$

- (b) Define resistance of a conductor and give its units. Name the factors on which the resistance of conductor at a fixed temperature depends. Write the corresponding relation between them.

- (b) Define resistance of a conductor and give its units. Name the factors on which the resistance of conductor at a fixed temperature depends. Write the corresponding relation between them.

CBSE 26

3

Resistance is defined as the opposition offered by conductor to the flow of current	$\frac{1}{2}$
Alternatively, $R = \frac{V}{I}$	
S.I unit: volt/ampere	
Alternatively, ohm (Ω)	$\frac{1}{2}$
Factors: 1) R is directly proportional to the length (l) of the conductor.	$\frac{1}{2}$
2) $R \propto \frac{1}{A}$, area of cross-section of the conductor.	$\frac{1}{2}$
3) Depends upon the nature of the conductor	$\frac{1}{2}$
Relation: $R = \rho \frac{l}{A}$	$\frac{1}{2}$

3. A copper wire is stretched to increase its length by 1%. Then the change in its resistance is close to

1

(A) 1%

(B) 4%

(C) -4%

(D) 2%

CBSE 26

3. A copper wire is stretched to increase its length by 1%. Then the change in its resistance is close to

1

(A) 1%

(B) 4%

(C) -4%

(D) 2%

CBSE 26

(D) 2%

6. A current of 1.5 A is maintained in a copper wire of length 1 m with area of cross-section $1.7 \times 10^{-7} \text{ m}^2$. The magnitude of electric field in the wire is $[\rho_{\text{Cu}} = 1.7 \times 10^{-8} \Omega \text{ m})$ 1

(A) $0.15 \frac{\text{V}}{\text{m}}$

(B) $0.30 \frac{\text{V}}{\text{m}}$

(C) $1.5 \frac{\text{V}}{\text{m}}$

(D) $3.0 \frac{\text{V}}{\text{m}}$

6. A current of 1.5 A is maintained in a copper wire of length 1 m with area of cross-section $1.7 \times 10^{-7} \text{ m}^2$. The magnitude of electric field in the wire is $[\rho_{\text{Cu}} = 1.7 \times 10^{-8} \Omega \text{ m})$

1

(A) $0.15 \frac{\text{V}}{\text{m}}$

(B) $0.30 \frac{\text{V}}{\text{m}}$

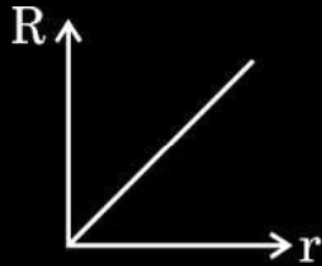
(C) $1.5 \frac{\text{V}}{\text{m}}$

(D) $3.0 \frac{\text{V}}{\text{m}}$

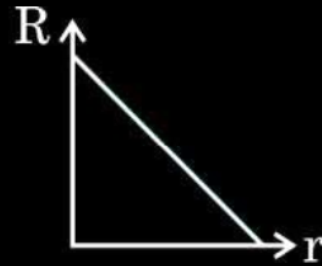
CBSE 26

(A) $0.15 \frac{\text{V}}{\text{m}}$

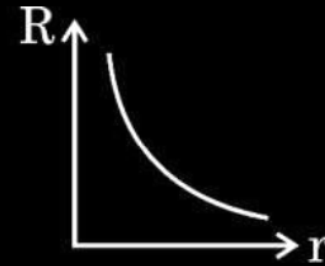
15. The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is :



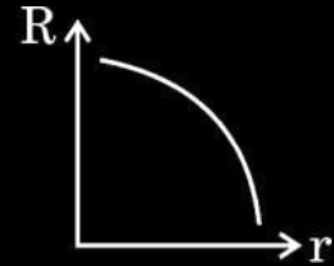
(a)



(b)

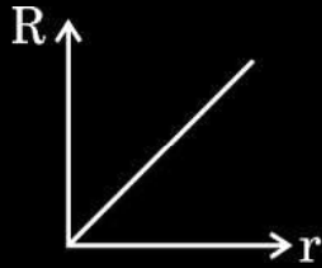


(c)

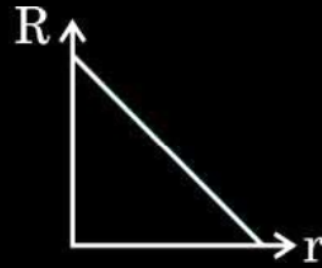


(d)

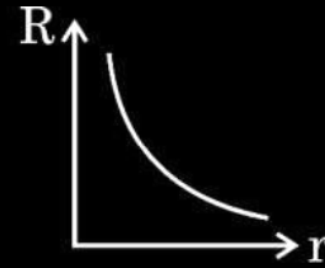
15. The correct graph showing the variation of the resistance (R) of a cylindrical metal wire as a function of its radius (r), keeping its length and temperature constant, is :



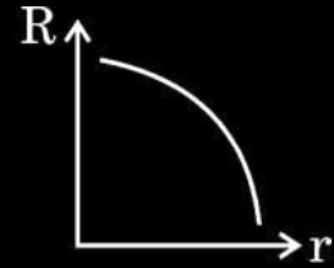
(a)



(b)



(c)



(d)



(e)

29. A potential difference V is applied across a conductor of length l and cross-sectional area A . Briefly explain how the current density j in the conductor will be affected if

(a) the potential difference V is doubled,

(b) the conductor were gradually stretched to reduce its cross-sectional area to $\frac{A}{2}$ and then the same potential difference V is applied across it.

3

(a) Effect of potential difference on current density	-	1 ½
(b) Effect of dimensions of conductor on current density	-	1 ½

(a) Current density $j = \frac{ne^2 E \tau}{m}$ $= \frac{ne^2 V \tau}{ml}$ Or $j \propto V$ If V is doubled, the current density will be doubled.	
(b) As the conductor were gradually stretched to reduce its cross sectional area to half ($A/2$), its length will become double ($2l$) as volume of conductor remains the same.	1
$\therefore j \propto \frac{1}{l}$	$\frac{1}{2}$
$\therefore$ If same potential difference is applied, the current density will become half.	$\frac{1}{2}$

17. An electric field E is maintained in a wire of length ' l ' and area of cross-section ' a '. Derive the relation between the current density ' σ ' in the wire and the electric field E .

17. An electric field E is maintained in a wire of length ' l ' and area of cross-section ' a '. Derive the relation between the current density ' σ ' in the wire and the electric field E .

2

Deriving relation		2
$V = IR$		$\frac{1}{2}$
$El = \frac{I \rho l}{A}$	$(V = El, R = \frac{\rho l}{A})$	$\frac{1}{2}$
$E = \frac{I}{A} \rho$		$\frac{1}{2}$
$E = \sigma \rho$		$\frac{1}{2}$

3. In a uniform straight wire, conduction electrons move along + x direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then :
- (A) $\vec{E}$ and $\vec{j}$ both are along + x direction.
 - (B) $\vec{E}$ and $\vec{j}$ both are along - x direction.
 - (C) $\vec{E}$ is along + x direction, but $\vec{j}$ is along - x direction.
 - (D) $\vec{E}$ is along - x direction, but $\vec{j}$ is along + x direction.

3. In a uniform straight wire, conduction electrons move along + x direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then :
- (A) $\vec{E}$ and $\vec{j}$ both are along + x direction.
 - (B) $\vec{E}$ and $\vec{j}$ both are along - x direction.
 - (C) $\vec{E}$ is along + x direction, but $\vec{j}$ is along - x direction.
 - (D) $\vec{E}$ is along - x direction, but $\vec{j}$ is along + x direction.

(B) $\vec{E}$ and $\vec{j}$ both are along -x direction

17. A uniform wire of length L and area of cross-section A has resistance R . The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire.

17. A uniform wire of length L and area of cross-section A has resistance R . The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire.

2

As the wire is uniformly stretched;	
$\Rightarrow Al = A'l' \Rightarrow A' = \frac{lA}{1.25l}$	$\frac{1}{2}$
$A' = \frac{4}{5}A$	
$R' = \frac{\rho l'}{A'} = \frac{\rho(1.25l)}{\frac{4}{5}A}$	
$R' = \frac{25}{16}R$	1
% increase = $\left(\frac{R' - R}{R}\right) \times 100$	
$= \left(\frac{25}{16} - 1\right) \times 100 = 56.25\%$	$\frac{1}{2}$

15. *Assertion (A) :* The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

15. *Assertion (A) :* The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

(iii) A wire of length 0.5 m and cross-sectional area $1.0 \times 10^{-7} \text{ m}^2$ is connected to a battery of 2 V that maintains a current of 1.5 A in it. The conductivity of the material of the wire (in $\Omega^{-1} \text{ m}^{-1}$) is : 1

(A) 2.5×10^4

(B) 3.0×10^5

(C) 3.75×10^6

(D) 5.0×10^7

(iii) A wire of length 0.5 m and cross-sectional area $1.0 \times 10^{-7} \text{ m}^2$ is connected to a battery of 2 V that maintains a current of 1.5 A in it. The conductivity of the material of the wire (in $\Omega^{-1} \text{ m}^{-1}$) is : 1

(A) 2.5×10^4

(B) 3.0×10^5

(C) 3.75×10^6

(D) 5.0×10^7

- (b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law $\vec{j} = \sigma \vec{E}$. **3**

Defining current density and Writing its SI unit	1+ ½
Deriving $\vec{j} = \sigma \vec{E}$	1 ½

Current density – It is defined as the current through unit area normal to the current.

Alternatively, $\vec{J} = \frac{I}{\vec{A}}$

S.I unit A/m²

Derivation

$$V = IR$$

$$\vec{E}l = I \frac{\rho l}{\vec{A}}$$

$$\vec{E} = \vec{J} \rho$$

$$\vec{J} = \frac{I}{\rho} \vec{E} = \sigma \vec{E}$$

1

1/2

1/2

1/2

1/2

18. A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes $40\ \Omega$, find the value of X .

18. A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes $40\ \Omega$, find the value of X .

2

$$R = n^2 x$$

$$40 = (2)^2 x$$

$$x = 10\ \Omega$$

Alternatively

Volume of wire before and after stretch will remain same

$$A_1 l_1 = A_2 l_2$$

$$A_2 = \frac{A l_1}{l_2}$$

$$= \frac{A}{2}$$

$$R = \rho \frac{l_2}{A_2}$$

$$40 = 4\rho \frac{2l}{A/2}$$

$$\therefore x = 10\Omega$$

$\frac{1}{2}$

$\frac{1}{2}$

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A , R_B and R_C respectively, then :

(A) $R_A > R_B > R_C$

(B) $R_B > R_C > R_A$

(C) $R_C > R_A > R_B$

(D) $R_A > R_C > R_B$

2. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A , R_B and R_C respectively, then :

(A) $R_A > R_B > R_C$

(B) $R_B > R_C > R_A$

(C) $R_C > R_A > R_B$

(D) $R_A > R_C > R_B$

2. A current flows through a cylindrical conductor of radius R . The current density at a point in the conductor is $j = \alpha r$ (along its axis), here α is a constant and r is distance from the axis of the conductor. The current flowing through the portion of the conductor from $r = 0$ to $r = \frac{R}{2}$ is proportional to :

(A) R

(B) R^2

(C) R^3

(D) R^4

2. A current flows through a cylindrical conductor of radius R . The current density at a point in the conductor is $j = \alpha r$ (along its axis), here α is a constant and r is distance from the axis of the conductor. The current flowing through the portion of the conductor from $r = 0$ to $r = \frac{R}{2}$ is proportional to :

(A) R

(B) R^2

(C) R^3

(D) R^4

21. (a) Two wires of the same material and the same radius have their lengths in the ratio $2 : 3$. They are connected in parallel to a battery which supplies a current of 15 A. Find the current through the wires. **2**

(a)

Finding current

2

$$R_1 = \frac{\rho l_1}{A}; R_2 = \frac{\rho l_2}{A}$$

$$\frac{l_1}{l_2} = \frac{2}{3} \Rightarrow \frac{R_1}{R_2} = \frac{2}{3}$$

$$I \propto \frac{1}{R}$$

$$\Rightarrow \frac{I_1}{I_2} = \frac{3}{2}$$

$$\Rightarrow I_1 = \frac{3}{5} \times 15 = 9A$$

$$\Rightarrow I_2 = \frac{2}{5} \times 15 = 6A$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

2. A student has three resistors, each of resistance R . To obtain a resistance of $\frac{2}{3}R$, she should connect

1

- (A) all the three resistors in series.
- (B) all the three resistors in parallel.
- (C) two resistors in series and then this combination in parallel with the third resistor.
- (D) two resistors in parallel and then this combination in series with the third resistor.

2. A student has three resistors, each of resistance R . To obtain a resistance of $\frac{2}{3}R$, she should connect

1

(A) all the three resistors in series.

(B) all the three resistors in parallel.

(C) two resistors in series and then this combination in parallel with the third resistor.

(D) two resistors in parallel and then this combination in series with the third resistor.

2. The resistance of a wire of length L and radius r is R . Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$? 1

- (A) Using a wire of same radius and twice the length
- (B) Using a wire of same radius and half length
- (C) Using a wire of same length and twice the radius
- (D) Using a wire of same length and half the radius

2. The resistance of a wire of length L and radius r is R . Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$? 1

(A) Using a wire of same radius and twice the length

(B) Using a wire of same radius and half length

(C) Using a wire of same length and twice the radius

(D) Using a wire of same length and half the radius

17. Show that $\vec{E} = -\rho \vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material.

17. Show that $\vec{E} = \rho \vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material.

2

Obtaining Ohm's law from $\vec{E} = \rho \vec{j}$	1 1/2	
Writing the condition	1/2	
$\vec{E} = \rho \vec{j}$		1/2
$\frac{V}{l} = \rho \frac{I}{A}$		1/2
$\frac{V}{I} = \rho \frac{l}{A} = \text{constant (Ohm's Law)}$		1/2
Condition for non- validity of Ohm's Law		
High temperature / in semiconductor		1/2

28. (a) Differentiate between electrical resistance and resistivity of a conductor.
- (b) Two metallic rods, each of length L , area of cross A_1 and A_2 , having resistivities ρ_1 and ρ_2 are connected in parallel across a d.c. battery. Obtain the expression for the effective resistivity of this combination.

(a) Electrical resistance (R) of a conductor equals the ratio of the potential difference (V) applied across it, to the resulting current (I) flowing through it. (Alternatively: $R = \frac{V}{I}$)

1

The resistivity of a conductor equals the resistance of a wire of unit length and unit area of cross section, drawn from the material of that conductor. (Alternatively: $R = \rho \frac{l}{A}$ or $\rho = \frac{RA}{l}$)

1

(or any other one relevant difference)

(b) For the parallel combination equivalent resistance is given by

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

1/2

$$\frac{A_1 + A_2}{\rho_{eq} L} = \frac{A_1}{\rho_1 L} + \frac{A_2}{\rho_2 L}$$

Where $(A_1 + A_2)$ is the effective area of cross section of combined rod in parallel combination of the rods.

$$\frac{\rho_1 \rho_2}{(\rho_2 A_1 + \rho_1 A_2)} = \frac{\rho_{eq}}{(A_1 + A_2)}$$
$$\Rightarrow \rho_{eq} = \frac{\rho_1 \rho_2 (A_1 + A_2)}{(\rho_2 A_1 + \rho_1 A_2)}$$

1/2

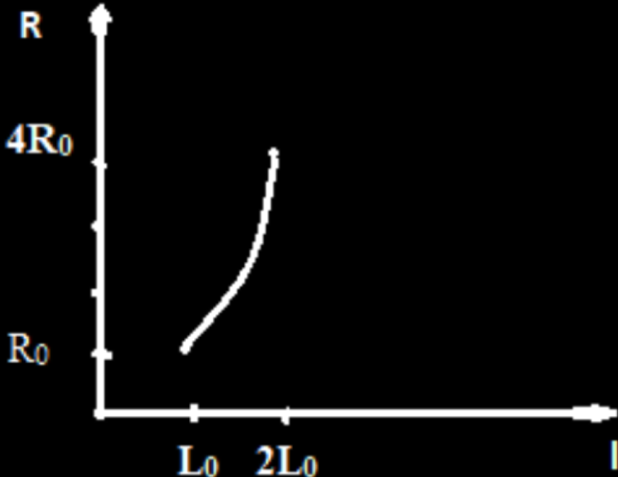
(Note : If a student just writes the expression of equivalent resistance, award half mark of this part)

22. A wire of length L_0 has a resistance R_0 . It is gradually stretched till its length becomes $2 L_0$.
- (a) Plot a graph showing variation of its resistance R with its length l during stretching.
 - (b) What will be its resistance when its length becomes $2 L_0$?

22. A wire of length L_0 has a resistance R_0 . It is gradually stretched till its length becomes $2L_0$.

- (a) Plot a graph showing variation of its resistance R with its length l during stretching.
- (b) What will be its resistance when its length becomes $2L_0$?

2

(a) Graph 1 (b) Final resistance 1	$\frac{1}{2}$
(a) $R = S \frac{l}{A} = S \frac{l^2}{V}$ 	$\frac{1}{2}$
[Note if a student draws only the graph correctly, award full 1 mark] (b) Resistance becomes $4R_0$ (Hint :- As $R = \frac{\rho l}{A} = \frac{\rho l^2}{Al} = \frac{\rho l^2}{V}$) (V= Volume) $R \propto l^2$	1

4. The electrical resistance of a conductor

1

- (A) varies directly proportional to its area of cross-section.
- (B) decreases with increase in its temperature.
- (C) decreases with increase in its conductivity.
- (D) is independent of its shape but depends only on its volume.

4. The electrical resistance of a conductor
- (A) varies directly proportional to its area of cross-section.
 - (B) decreases with increase in its temperature.
 - (C) decreases with increase in its conductivity.
 - (D) is independent of its shape but depends only on its volume.

2. In a current carrying conductor, the ratio of the electric field and the current density at a point is called

1

- (A) Resistivity
- (B) Conductivity
- (C) Resistance
- (D) Mobility

2. In a current carrying conductor, the ratio of the electric field and the current density at a point is called

1

(A) Resistivity

(B) Conductivity

(C) Resistance

(D) Mobility